QQ-P-416a
DECEMBER 11, 1956
SUPERSEDING
Fed. Spec. QQ–P–416
September 12, 1949

FEDERAL SPECIFICATION

PLATING, CADMIUM (ELECTRODEPOSITED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Classes.—Cadmium plating covered by this specification shall be of the following classes, as specified in the invitation for bids:

Class 1.—0.00050 inch thick.
Class 2.—0.00030 inch thick.
Class 3.—0.00020 inch thick.

1.2 Types.—The cadmium plating shall be of the following types, as specified in the invitation for bids:

Type I.—Without supplementary chromate or phosphate treatment.
Type II.—With supplementary chromate treatment (see 3.3.2 and 6.1.1).
Type III.—With supplementary phosphate treatment (see 3.3.3 and 6.1.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following Federal Test Method Standard of the issue in effect on date of invitation for bids, forms a part of this specification:

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Specifications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Materials.—The materials used shall be such as to produce coatings which meet the requirements of this specification.

3.2 Workmanship.

3.2.1 *Basis metal.*—The basis metal shall be substantially free from defects that will be detrimental to the appearance or the protective value of the plating. It shall be subjected to such cleaning, pickling, and plating procedures as are necessary to yield deposits as hereinafter specified.

3.2.2 *Plating.*—The cadmium deposit shall be smooth, fine-grained, adherent, and free from blisters, pits, nodules, indications of burning, and other defects. Superficial stain-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

ing which has been demonstrated as resulting from rinsing, or slight discoloration resulting from baking as specified below, shall not be cause for rejection. All details of workmanship shall conform to the best practice for high quality plating.

3.3 General requirements.

3.3.1 Unless otherwise specified, the plating shall be applied after all machining, brazing, welding, forming, and perforating of the article have been completed.

3.3.2 Unless otherwise specified, the chromate treatment required for conversion to type II shall be either a chemical or electrochemical treatment in an aqueous solution of salts, acids, or both, and containing hexavalent chromium and activating acid radical(s) to produce a continuous smooth, distinct protective film, distinctly colored iridescent bronze to brown including olive drab and yellow. The articles so treated shall be thoroughly rinsed and dried according to requirements for the particular chemical or electrochemical process used. Type II plating shall be similar in appearance to coatings on separate specimens which are capable of passing the salt-spray test. (See 3.4.2, 4.3.3.2, and 4.4.3.) The ordinary chromic and nitric acid bright dips for cadmium are not intended as chromate treatments.

3.3.3 Unless otherwise specified, the phosphate treatment required for conversion to type III shall be a chemical treatment with a suitable aqueous solution containing phosphoric acid, phosphates and accelerating chemicals to produce a tightly adherent film. (See 3.4.3.)

3.3.4 All ferrous parts having a hardness greater than Rockwell C–40 shall be given a suitable heat treatment for stress relief prior to cleaning and plating if they contain objectionable residual stresses. (SEE 6.12)

3.3.5 Springs or other parts subject to flexure or repeated impact having a hardness greater than Rockwell C–40 shall be heated at 190° ± 14°C. (375° ± 25°F.) for 3 hours after plating. Hardened parts, 220,000 p.s.i. and above, and carburized parts, shall not be heated as above, but shall be treated by a method approved for such parts by the contracting agency. Plated springs shall not be flexed prior to the baking operation. In the case of type II and type III parts, the heat treatment shall be given prior to the application of the supplementary coatings. (See 6.7.)

3.3.6 Cadmium shall be deposited directly on the basis metal without a preliminary plating of other metal, such as nickel, except in the case of parts made of corrosion-resisting steel on which a preliminary plating of nickel is permissible.

3.3.7 When aluminum is the basis metal the cadmium plating process shall be as specified by the procuring agency.

3.3.8 Unless otherwise specified, either a bright or dull finish will be acceptable.

3.4 Detail requirements.

3.4.1 *Thickness of plating.*—Unless otherwise specified, the thickness of cadmium shall be as specified in table I on all visible surfaces which can be touched by a ball 0.75 inch in diameter.

TABLE I.—*Thickness*

Class	Thickness, minimum
	Inch
1	0.00050
2	.00030
3	.00020

3.4.2 *Salt-spray requirements.* — Type II plating shall show no white corrosion products of cadmium as defined in 3.4.2.1 at the end of the periods of time shown in table II when tested by continuous exposure to the salt spray in accordance with 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE II.—*Salt-spray test*

Type	Class	Test period for white corrosion products
		Hours
Type II	Class 1	96
	Class 2	96
	Class 3	96

3.4.2.1 For type II plating, the appearance of white corrosion products, visible to the unaided eye at normal reading distance, at accidental scratches through the chromate film to the plate or at unscratched areas of the chromate film shall be cause for rejection, except that white corrosion products at edges shall not constitute failure.

3.4.3 The supplementary phosphate treatment (type III) shall afford a good base for paint. Paint coatings applied to cadmium surfaces with the supplementary phosphate coatings shall show no blistering following subjection to the water-resistance test described in 4.4.4.

3.4.4 *Adhesion.*—The adhesion of the plating shall be such that when examined at a magnification of approximately 4 diameters, it does not show separation from the basis metal at the interface when subjected to the tests described in 4.4.2. The interface between the cadmium and the basis metal is the surface of the basis metal before plating. The formation of cracks in the plate caused by rupture of the basis metal which do not result in flaking, peeling, or blistering of the plate shall not be considered as nonconformance to this requirement.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Place of Inspection and tests.—Unless otherwise specified, sampling, inspection, and tests shall be made at the place of manufacture.

4.2 Separate specimens.—When the plated articles are of such form as to be not readily adaptable to a test specified herein, or at the discretion of the inspector for destructive tests for the sampling of small lot sizes, the test shall be made by the use of separate specimens plated concurrently with the articles represented. The separate specimens shall be of a basis metal equivalent to that of the articles represented. "Equivalent" basis metal includes chemical composition and finish of surface prior to plating. For example, a cold-rolled steel surface should not be used to represent a hot-rolled steel surface; due to the impracticality of forging or casting separate test specimens, hot-rolled steel specimens may be used to represent forged and cast-steel articles. The separate specimens shall be strips approximately 1 inch wide, 4 inches long and 0.04 inch thick, for adhesion tests; but shall be at least 4 inches wide, 6 inches long and approximately 0.04 inch thick for all other tests. These specimens shall be introduced into a lot at regular intervals prior to the cleaning operations preliminary to plating and shall not be separated therefrom until after completion of the processing. Conditions affecting the plating of the specimens including the spacing and positioning with respect to anodes and to other objects being plated shall correspond as nearly as possible to those affecting the significant surfaces of the articles represented. *Separate specimens shall not be used for thickness measurements, however, unless the necessity for their use has been demonstrated.*

4.3 Sampling.

4.3.1 *Lot.* — A lot shall consist of plated articles of approximately the same size, shape, type, and class of plating submitted for inspection at one time.

4.3.2 *Sampling procedure for visual inspection and non-destructive tests for coating thickness.*—A sample shall be drawn from each inspection lot by taking at random from the lot the number of articles indicated in table III. The lot shall be accepted or rejected according to the procedures in 4.3.2.1 and 4.3.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE III.—*Sampling for visual inspection and non-destructive tests of coating thickness*

Number of items in inspection lot	Number of items in samples (randomly selected)	Acceptance number (maximum number of sample items nonconforming to any test).
15 or less	*7	0
16 to 40	10	0
41 to 110	15	0
111 to 300	25	1
301 to 500	35	1
501 and over	50	2

* If the number of items in the inspection lot is less than 7, the number of items in the sample shall equal the number of items in the inspection lot.

4.3.2.1 *Visual Inspection.*—The number of sample articles indicated in table III shall be taken at random from the lot to be inspected and compared with requirements of 3.2 and 3.3.2 after plating. If the number of nonconforming articles exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2.2 *Thickness (nondestructive tests).*—The number of articles indicated in table III shall be taken from the lot to be inspected and the plating thickness measured at several locations on each article where the plating would be expected to be minimum. The article shall be considered to be nonconforming if one or more measurements fail to meet the specified minimum thickness. If the number of nonconforming articles in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.3 *Sampling for destructive tests of coating thickness, adhesion, salt spray, and water resistance.*—A random sample of four articles shall be taken from each inspection lot ~~by the Government inspector~~ or four separately plated specimens shall be prepared in accordance with 4.2 to represent each inspection lot. If the number of the articles in the inspection lot is four or less, the number of the articles in the sample shall be determined by the procuring agency or shall be left to the discretion of the inspector.

4.3.3.1 *Thickness (destructive tests).* — Each article shall be tested in several locations at which the plating would be expected to be minimum. If the plating thickness at any place on any article is less than the specified thickness, the lot shall be rejected.

4.3.3.2 *Salt spray.*—For type II coatings, a set of 4 separate specimens, in lieu of the articles, shall be tested. These tests shall be made periodically at the discretion of the inspector or when there is reason to believe that the type II supplementary treatment solution might not be operating properly.

4.3.3.3 *Adhesion.* — The articles used for the thickness tests, if of suitable size and form, may be used as specimens for the adhesion tests. Failure of one or more of the test specimens shall constitute failure of the lot.

4.3.3.4 *Water resistance.* — The right is reserved to subject to the water resistance test described in 4.4.4 any cadmium-plated articles drawn at random from the lot to be inspected, or separate test specimens prepared as described in 4.2. Failure of one or more of the test specimens shall constitute failure of the lot.

4.4 Test methods.

4.4.1 *Thickness.* — Thickness determinations shall be made by the drop test or microscopic test methods described herein or by instrument methods such as a magnetic method of a type approved by the procuring agency, except that for referee tests the microscopic method shall be used (see 6.3). Measurement of minimum plating thickness on bolts shall be made on the shank adjacent to the thread, or on the head, if the shank is completely threaded. Thickness measurements of type II and type III cadmium shall be made after the application of type II and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

type III supplementary coatings and subsequent removal. The chromate film may be removed from type II plating by using a very mild abrasive (a paste of levigated alumina rubbed on with the finger). The phosphate coating may be removed from type III plating by scrubbing with a rubber policeman in a 10 percent solution of NaOH (usually takes from 10 to 15 minutes).

4.4.1.1 *Drop test.*

4.4.1.1.1 *Apparatus.*—The apparatus shall consist of a 250-milliliter separatory funnel having an outlet equipped with one stopcock and a capillary tube approximately 0.025 inch (0.63 mm.) inside bore and 5.5 inches (14 cm.) long. The lower end of the capillary tube shall be drawn to a small orifice, the drops from which are about 0.05 milliliter each. To preserve an effectively constant level, a small glass tube shall be inserted into the top of the funnel through a rubber stopper, and its position so adjusted that when the stopcock is open the desired rate of flow from the orifice shall be obtained.

4.4.1.1.2 *Test solution.*—The test solution shall be as follows:

Chromic acid (C.P.) ----------	200 grams
Sulfuric acid (C.P.), sp.gr. 1.84	27 milliliters
Distilled water to make --------	1 liter

4.4.1.1.3 *Procedure.* — The plated article and the solution shall be kept in the test room long enough to acquire the temperature of the room which shall be within the range shown in table IV. The temperature shall be noted and recorded to within 2°F. (1°C.). The funnel shall be filled with the test solution and the rate of flow from the orifice adjusted to not less than 95 nor more than 105 drops per minute. Immediately before testing, the plated article shall be cleaned with an organic solvent, followed by light rubbing with pure magnesium oxide paste, rinsed in water, dried and brought to room temperature. The surface to be tested shall be considered to be sufficiently clean when the water rinse shows freedom from "water break," that is, when water wets the entire surface. The cleaned article shall be supported by a rack, clasp, or other suitable means so that the test surface is at an angle of about 45° from the horizontal and from 0.7 to 1 inch below the orifice. The stopcock then shall be opened wide and the elapsed time, in seconds, from the time the first drop strikes the surface until the basis metal is first revealed is determined with a stop watch. The endpoint is the first appearance of the basis metal, recognizable by the change in color at that point. The spent solution shall not be used for subsequent tests. The plating thickness shall be determined from table IV.

4.4.1.2 *Microscopic test.* — The specimen shall consist of a cross section of a significant surface suitably mounted and ground and polished. The polished surface shall be etched with a suitable reagent (see 6.3) to differentiate the cadmium plating from the basis metal. The thickness of coating at any desired point on the exposed section shall be measured by either of the methods described below. Specimens may be mounted in a plastic material such as a phenolic or acrylate resin. Flat specimens may be either mounted in pairs, coating-to-coating, or in clamps. Specimens that are not flat may be mounted against a piece of brass or steel conforming to the shape of the specimen. In mounting, care should be taken that voids do not form between the specimen and the mounting material. The specimen shall be held rigidly and in such a position that the surface tested is perpendicular to the flat face that is to be ground and polished (a deviation of 10° from normal introduces an error of about 2 percent in thickness) or at a known angle, in which case, the thickness measured must be corrected accordingly. The pieces may be plated first with a relatively thick coating of copper to protect the edges during grinding and polishing. Cleaning of the specimens prior to plating should not involve strongly acid or alkaline solutions or abrasives as these may remove a significant amount of the cadmium coating. The first layer of copper shall be deposited from a cyanide solution, after

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE IV.—*Time-temperature-plating thickness relations*

Time, seconds	Plating thickness, inches						
	65°F.	*70°F.*	*75°F.*	*80°F.*	*85°F.*	*90°F.*	*95°F*
8	0.00010	0.00010	0.00011	0.00011	0.00012	0.00012	0.00013
10	.00013	.00013	.00014	.00014	.00014	.00015	.00016
12	.00016	.00016	.00016	.00017	.00017	.00018	.00019
14	.00018	.00018	.00019	.00019	.00020	.00021	.00022
16	.00021	.00021	.00022	.00022	.00023	.00024	.00025
18	.00023	.00024	.00025	.00025	.00026	.00027	.00028
20	.00026	.00027	.00027	.00028	.00029	.00030	.00032
22	.00029	.00029	.00030	.00030	.00031	.00033	.00035
24	.00031	.00032	.00033	.00033	.00034	.00036	.00038
26	.00034	.00034	.00035	.00036	.00037	.00039	.00041
28	.00037	.00037	.00038	.00039	.00040	.00042	.00044
30	.00039	.00040	.00041	.00042	.00043	.00045	.00047
32	.00042	.00043	.00044	.00045	.00046	.00048	.00051
34	.00044	.00045	.00046	.00048	.00049	.00051	.00054
36	.00047	.00048	.00049	.00050	.00051	.00054	.00057
38	.00050	.00051	.00052	.00053	.00054	.00057	.00060
40	.00052	.00053	.00055	.00056	.00057	.00060	.00063
42	.00055	.00056	.00057	.00058	.00060	.00063	.00066
44	.00057	.00059	.00060	.00061	.00063	.00066	.00070
46	.00060	.00062	.00063	.00064	.00066	.00069	.00073
48	.00063	.00064	.00065	.00067	.00069	.00072	.00076

which, if desired, an acid copper solution may be used. The section to be examined shall be ground and polished by regular metallographic methods, using successively finer abrasives, the last of which shall not be coarser than 500 mesh.

4.4.1.2.1 The thickness shall be measured with a filar micrometer ocular that has been calibrated against a standard scale. The accuracy of the calibration shall be within plus or minus 2 percent; or the method of measuring shall be as specified in 4.4.1.2.2.

4.4.1.2.2 The image of the specimen shall be projected at a known magnification on the ground-glass focusing plane of the camera of a metallographic microscope. The width of the projected line of deposit, when measured with a graduated linear scale, and divided by the magnification (for example, 500), gives the thickness of the coating.

4.4.2 *Adhesion.*—Adhesion may be determined by scraping the surface of the plated article to expose the basis metal and examining at four diameters magnification for evidence of nonadhesion. Alternatively, the article may be clamped in a vice and the projecting portion bent back and forth until rupture occurs. If the edge of the ruptured plate can be peeled back or if separation between the plate and the basis metal can be seen at the point of rupture when examined at four diameters magnification adhesion is not satisfactory.

4.4.3 *Salt-spray test.*—The salt-spray test shall be made in accordance with Federal Test Method Standard No. 151. To secure uniformity of results, type II supplementary coatings shall be aged for 24 hours before subjection to the salt spray.

4.4.4 *Water-resistance test.*—The samples, after receiving the supplementary phosphate (type III) treatment with the lot represented shall be coated with 0.7 to 1.2 mils (dryfilm thickness) of the primer or enamel intended for use on the article. The coated specimens

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

shall be dried in conformance with the requirements of the coating material, either by air drying for 90 hours or by baking at the required temperature for the required period of time followed by 24 hours air drying. The specimens shall then be immersed in distilled water at pH 5.0 to 7.0 at 75°F., minimum, for 24 hours. The appearance of blisters in the paint coating shall be cause for rejection.

4.4.5 *Embrittlement relief.*—Inspection for determination of compliance with 3.3.5 shall be made as specified by the procuring agency. (See also 7.3.5.)

4.5 **Retests.**—Plated articles which have been subjected to the drop test and articles rejected or withdrawn because of defective or deficient plating may be resubmitted after stripping and replating or screening of the entire lot. Full particulars concerning the replating shall be furnished to the inspector. Where only insufficient thickness is involved, replating without stripping the existing coating may be permitted at the discretion of the inspector. All material having a hardness greater than Rockwell C–40 shall be hydrogen-embrittlement relieved after plating.

5. PREPARATION FOR DELIVERY

5.1 There are no packaging, packing, or marking requirements applicable to this specification. (See 6.9.)

6. NOTES

6.1 Intended use.

6.1.1 The primary purpose of chromate finishes is to retard or prevent the formation of white corrosion products on surfaces exposed to stagnent water, high humidity atmospheres, salt water, marine atmospheres, or exposed to cyclic condensation and drying. Some types of chromate coatings have proved satisfactory as a base for paint.

6.1.2 The primary purpose of phosphate finishes is to form a paint base.

6.2 **Ordering data.** — Purchasers should specify the type and class of plating required and the method of determining compliance with the requirements of 3.3.5 and 4.2, and should exercise any desired options offered herein. (See 1.1, 1.2, 3.3.1, 3.3.2, 3.3.3, 3.3.7, 3.3.8, 3.4.1, 4.1, 4.2, and 4.4.5.)

6.3 The following typical reagent has been satisfactory for etching cadmium plating on steel for microscopic determination of plating thickness:

Chromic acid	200 grams per liter
Sodium sulfate	15 grams per liter

6.4 **Toxicity.** — Cadmium, because of its toxicity, should not be employed as a coating for any object intended for use as a food container or cooking utensil or for any object likely to come in contact with food or beverages. Cadmium-coated sheets and any other structural shapes which may be subjected to heat from welding or soldering operations should be so labeled because of the danger from poisonous vapors during such operations.

6.5 **Type II.**—In general, supplementary chromate treatments (not to be painted) are damaged if used on cadmium-plated parts which are continuously exposed to temperatures in excess of 150°F. or intermittently exposed for short periods to temperatures of approximately 300°F.

6.6 Supplementary chromate treatments (type II) which involve only dipping in chemical solutions normally require a sufficient period of drying to render the parts suitable for handling without damage to the coating while in gelatinous form; and it is important with such that the coating is not excessively damaged while wet. Chromating processes of the electrochemical type normally produce conversion films that do not require drying to avoid film fracture in production.

6.7 Surfaces of cadmium plating intended for conversion to type II and which have become passive as a result of the heat treat-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

ment specified in 3.3.5 may be reactivated by brief immersion in dilute acid. If, for example, the chromating solution is acidified with sulphuric acid; then the reactivating solution should be 1 part sulphuric acid (sp. gr. 1.83) by volume added to 99 parts water, or for further example; if the chromating solution is acidified with hydrochloric acid; then the activating solution should be 1 part hydrochloric acid (sp. gr. 1.16) by volume added to 99 parts water. The duration of immersion should be as brief as is consistent with the nature of the work. For example, a perforated container of barrel plated parts would be expected to be reactivated in approximately 15 seconds and separately racked items in approximately 5 seconds. The surfaces should be activated as soon as possible following heat treatment and should be handled carefully to avoid contamination by dirt or grease.

6.8 The dimensional tolerance of most threaded articles, such as nuts, bolts, and similar fasteners with complementary threads do not permit the application of a coating thickness much greater than class 3. If heavier coatings are required for satisfactory corrosion resistance, allowances must be made in the manufacture of the threaded fasteners for tolerance necessary for plate build-up.

6.9 Unprotected cadmium-plated articles should not be packaged in non-ventilated containers, either together or in contact with electrical equipment, because of the danger of deleterious effects on cadmium plating from unstable organic electrical insulation. In addition to organic electrical insulation, phenolic resinous substances and others containing unsaturated carbon to carbon linkages, such as oil paints and impregnated paper, etc., cause an abnormal attack on cadmium by setting free in the presence of moisture formic acid, butyric acid, etc. Corrosion of cadmium coatings and the steel basis metal has been noted when cadmium plated articles have been packaged in direct contact with container materials such as wood or cardboard. Corrosion has been especially severe if the container materials have become wet or have been stored under conditions of high humidity.

6.10 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply to all other respects.

6.11 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Air Force, Army and Bureau of Aeronautics.

7.2.1 The equipment and processes employed to accomplish cadmium plating as required by this specification shall be subject

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

to the approval of an authorized Government inspector.

7.2.2 Unless otherwise specified, cadmium plating supplied for aeronautical use, shall be type 1 and the thickness requirements shall be class 2 unless otherwise specified or controlled by the following exceptions:

1. Bolts, studs, washers, and articles with portions externally threaded. Those parts shall have a minimum of class 3 thickness.
2. Parts whose dimensional tolerances will not permit a class 2 thickness shall be given a maximum thickness compatible with dimensional tolerances.
3. Holes, recesses, internal threads and other areas where a controlled deposit cannot be normally obtained shall not be subject to a thickness requirement.

7.3 Navy.

7.3.1 *Applicable specification.*

7.3.1.1 *Navy Department specification:*

General Specifications for Inspection of Material, and Appendix II, Metals—Part A, Definitions and Tests (Part A is applicable, except for Bureau of Aeronautics, in lieu of Federal Test Method Standard No. 151 referred to in 2.1).

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3.2 *Inspection.* — The general inspection procedures shall be in accordance with Navy Department General Specification for Inspection of Material.

7.3.3 *For bureaus other than the Bureau of Aeronautics.*—Unless otherwise specified, a representative sample shall be taken by the Inspector according to the sampling procedure specified herein and shall be forwarded to a Government laboratory for tests to determine compliance with this specification.

7.3.4 For Bureau of Ordnance and Bureau of Ships only.

7.3.4.1 Salt-spray testing of type II coatings submitted for lot acceptance and inspection shall be conducted on plated parts wherever practicable, in lieu of separate panels, as specified in 3.4.2, 4.2, and 4.3.3.2.

7.3.5 *For Bureau of Ordnance only.* — In cases where hydrogen embrittlement may have occurred in the replating of springs or other parts subject to flexure, the acceptability of replated material shall be referred to the procuring activity.

MILITARY INTERESTS:
Army—O C Q E M Sig
Navy—Or A Sh MC
Air Force.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.